The cold-start insight for direct air capture survives exergy accounting of the model's heats and works, but its size, and with full heat recovery its existence, depend on pricing the chilling of the feed air

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AI-assistance disclosure. This preprint, its Lean code, its Python code and its claim ledger were produced by AI agents (Claude, Anthropic) under the author's direction. The author is responsible for the content. No human has audited the Lean statements for adequacy, and no human referee has checked the hand arguments. **Audit status (v2).** Version 1 was audited on 2026-10-27 by an independent AI agent (producer ≠ verifier; not a person), verdict ACCEPT_WITH_CORRECTIONS with four blocking items (`AUDIT.md`); this version 2 is the producer's revision (`REVISION.md`). The auditor recompiled the Lean core (rc 0) and replayed `Programmes.DacExergy.Core` with `leanchecker` (rc 0, 118 s; `data/leanchecker_replay_audit.json`). **The new scenario family of v2 (feed-air pricing, Section 3.5) was computed by the producer only**: where it overlaps the auditor's own numbers (three rows) they agree to the tolerances stated in Section 3.5, but an independent re-check of the family as a whole is pending. No statement lock exists. The source authors (NIST) have not been contacted and have not seen this work.

Abstract

McDannald et al. (arXiv 2501.04825) define an intrinsic capture efficiency $\xi = \Delta n_{CO_2}/E_{\text{total}}$ for a sorbent in an idealised piston cycle and conclude that it is "greatly beneficial to start cold": 11 of the 14 Pareto rows of their Table 2 start at $T_1 \leq 217.7$ K, and all start at or below 236.6 K. Their E_{total} is a face-value sum of heats and works: the cooling of Step 1 is counted as a heat of adsorption at T_1, the return Step 3 is set to zero (eq. 5), and no ambient temperature appears. We asked whether the conclusion survives when every heat of the sorbent and its adsorbate is weighted by its Carnot factor against an explicit ambient T_0, their sub-ambient cooling is priced, and the return step is charged; version 2 also prices the chilling of the contacted *feed air* ($1/x = 2{,}500$ mol of air per mol CO_2 at capture fraction 1), which v1 left out. **Our preregistered hypothesis H1 was that the optimum would move to $T_1 \geq T_0$ (prior 0.35). At the preregistered setting ($T_0 = 298.15$ K, no heat recovery) it is refuted with and without feed-air pricing.** (i) *Base accounting (feed air not priced, ideal machines):* for all 14 Table 2 rows the exergy-optimal start, found on a joint grid over T_1, ΔT and P_{end}, lies at 200–236.75 K in all 9 unconstrained variants ($T_0 \in \{273.15, 298.15, 313.15\}$ K, $r \in \{0, 0.5, 1\}$ heat recovery); the 9 purity-constrained variants are vacuous for every row (no warm path on our grid reaches the stored purity), so 126 of the 252 row-variants are informative, not 252. The cold-start benefit ratio (best ξ cold over best ξ with $T_1 \geq 298.15$ K) is 10–818 at face value, 3.5–129 at $r = 0$ and 1.29–6.6 at $r = 1$. (ii) *Feed air priced, no recuperation ($\varepsilon = 0$), ideal machines, $T_0 = 298.15$ K:* at $r = 0$ a cold start still wins in 14/14 rows, but much less cold ($T_1^* = 254$–288 K instead of 200–237 K) and by a much smaller factor (1.46–8.0); at $r = 0.5$, $T_1^* = 268$–290 K; at $r = 1$, **in 11 of 14 rows the optimum moves to ambient**: $T_1^* = 297.25$–297.75 K on a 0.25 K grid, and the best start below ambient beats the best start at or above it by only 0.07–0.35%, less than each row's own printed ξ uncertainty (not a paired test); a start ≥ 10 K below ambient then loses unless a recuperator recovers more than 91–96% of the air's cold (auditor's coarser split: 77–90%). Only VANNIK (both rows) and LETQAE01 keep a cold start (284–288 K) at every ε. So *the cold-start conclusion survives exergy accounting of the heats and works as published, but its size, and at full recovery its existence, depend on how the cost of chilling the feed air is priced.* Exergy-weighted efficiencies on the stored paths are 25.4–36.8 μmol/J against 5.98–11.26 at face value (nominal). **H2:** no published ξ exceeds $1/W_{\min}$ of the ideal mixture (dataset

maximum 11.259 μmol/J against 51.6). **Retraction:** v1 also reported five dataset entries with heat flows inconsistent with the second law; those flags were produced by our own Step-3 quadrature and vanish, over all 15,161 paths, when we use the published Step-2 rule. The revised numbers were computed by the producer; the auditor's own numbers agree where they overlap (three rows, within stated tolerances), and an independent re-check of the new feed-air scenario family is pending.

Tier split. [A] (Lean 4.34.1, axioms within {`propext`, `Classical.choice`, `Quot.sound`}): the twelve theorems of `DacExergy/Core.lean`, which are real-number algebra (Carnot weights in $(0, 1)$, a factorisation, sign and threshold equivalences, positivity and monotonicity of $RT_0 \ln(1/x)$; three are bare identities); they certify no number of the paper. The planted controls are caught by the axiom audit; the auditor's recompile and `leanchecker` replay pass (rc 0). No statement lock. [L]: the source paper's model (eqs. 1–10, Table 2) and the dataset and code licences, taken as read. [C]: the closed-form survival conditions (an N_2 threshold and a heat-capacity threshold) and mpmath-interval enclosures, about a *stylised two-component surrogate only*. [X]: the full-model re-accounting, the grid scans and scenario families, the sample, and the dataset-wide checks; the headline is Tier X. Every claim in the text carries one of these tags and a ledger identifier (`DAC-A-nnnn` etc.).

What this is NOT. These are thermodynamic limits of a published *simplified model* (idealised piston, Henry/IAST loadings, binary CO_2/N_2 "air", no water, no heat recovery in the original). It is **not** a statement about real plants, costs, deployment, sorbent performance in practice, or any climate effect, and it makes **no recommendation**. It is **not** an unconditional "cold start survives": the base accounting omits the chilling of the feed air and assumes ideal (Carnot) machines, and once the air is priced the result depends on the recuperator effectiveness ε, the heat-recovery bookend r and how cold "cold" is taken to be. The feed-air term assumes capture fraction 1 (the case most favourable to a cold start); r, ε and η are bookends, not designs. It does not say the NIST metric or its authors are wrong: face value and exergy are two accounting conventions and the source states its own scope. Nothing here has been discussed with the source authors.

Markers and ledger. [A]: kernel-checked Lean, axioms within the allowlist, controls caught; the tag attaches to the Lean statement, never to its physical gloss. [L]: a cited source taken as stated. [C]: executed interval code with a written argument, not machine-checked, *about the surrogate*. [X]: floating-point computation on the real model data; it supports nothing by itself beyond what it shows. Identifiers `DAC-A-nnnn`, `DAC-L-nnnn`, `DAC-C-nnnn`, `DAC-X-nnnn` refer to `ledger/ledger.json` (Section 8).

1 Setting, source and preregistration

This is item K1 of a computational programme (Proposal 4, preregistered 2026-10-27, `results/proposal4/2026-10-27/preregistration.json`). The source is McDannald, Siderius, DeCost, Choudhary and Ortiz-Montalvo, *Intrinsic Direct Air Capture* [1] (arXiv:2501.04825v1, 8 Jan 2025; the only version on arXiv), with its data [2] (Zenodo 10.5281/zenodo.14452152, from the NIST Data Publication 10.18434/mds2-3687, CC-BY-4.0) and its Julia code [3] (NIST IntrinsicDACCycle v0.0.2, NIST software licence). We credit the authors and NIST fully: every number below that is not labelled ours is theirs, and the headline of their paper is a thermodynamically careful model whose scope (no heat recovery, intrinsic comparison of sorbents) they state themselves. We read the paper in full (28 pages) and the relevant Julia sources line by line. **The supplement was not obtainable** (not on arXiv, not in the Zenodo record, not in the code repository) and was not read.

The preregistered item asked: (K1-H1) under exergy-consistent accounting the Pareto-optimal start temperature moves to $T_1 \geq T_0$ for most materials (prior 0.35); (K1-H2) no dataset entry has exergy-weighted efficiency above $1/W_{\min}$, and any violation flags a model-input inconsistency (definite answer, prior 0.45); (K1c) the text value "1.0×10^5 mol/J" versus Table 2 in μmol/J; (K1x) sensitivity to T_0 and heat recovery. Kill rules: if the dataset and code lacked per-path temperature-resolved energy terms, H1 would stay a Tier C argument; if the (unread) supplement showed that sub-ambient cooling is already priced, H1 would be withdrawn. Neither fired: the dataset exposes per-path arrays, and in the *code* the Step 3 term is $E_3 = 0$ in all four evaluators. The supplement being unread, we cannot say it is not priced elsewhere in the paper's text; the eq. 5 text says it is zero. Outcome (v2): H1 refuted at the preregistered setting ($T_0 = 298.15$ K, $r = 0$: 0/14 rows with $T_1^* \geq T_0$) in the base accounting *and* with the feed air priced; outside that setting, with the feed air priced and full

recovery ($r = 1$), the optimum moves to ambient (within 0.9 K) for 11/14 rows (Section 3.5). H2 answered (the published ξ never exceeds $1/W_{\min}$); the v1 "five inconsistent entries" are retracted (Section 3.6).

2 A step-by-step account for non-specialists

Step 1: the cycle as published. [L] A sorbent (here a metal-organic framework, MOF) sits in a piston. *Step 1:* it is exposed to "air" (400 ppm CO_2, balance N_2) at temperature T_1 and pressure P_1 and reaches equilibrium, releasing heat of adsorption that is carried away to keep T_1 fixed. *Step 2:* the chamber is closed and the temperature is raised and the pressure lowered along a path to $(T_{\text{end}}, P_{\text{end}})$; gas desorbs and leaves through a check-valve. *Step 3:* the system returns to (T_1, P_1). The captured amounts are Δn_{CO_2}, Δn_{N_2} (mol/kg), the purity is $x_{\text{end}} = \Delta n_{\text{CO}_2}/(\Delta n_{\text{CO}_2} + \Delta n_{\text{N}_2})$ (eq. 7) and the metric is

$$\xi = \frac{\Delta n_{\text{CO}_2}}{E_{\text{total}}} \quad \text{(eq. 8)}, \qquad E_{\text{total}} = E_1 + E_2 + E_3 \quad \text{(eq. 6)}.$$

Step 2: the energy balance as read. [L] Eq. 1: $E_1 = Q_{\text{ads}} - W_1$, which in the text becomes $E_1 = \Delta n_{\text{CO}_2}\big(q_{\text{ads,CO}_2}(T_1) + RT_1\big) + \Delta n_{\text{N}_2}\big(q_{\text{ads,N}_2}(T_1) + RT_1\big)$: a heat removed at T_1 plus a compression work. Eq. 2: $E_2 = Q_{\text{ads,CO}_2} + Q_{\text{ads,N}_2} - W_{2,\text{CO}_2} - W_{2,\text{N}_2} + E_{\text{heat,CO}_2} + E_{\text{heat,N}_2} + E_{\text{heat,sorb}} + E_{\Delta P}$, and eq. 4 writes it as an integral over a path s with $dT/ds \geq 0$, $dP/ds \leq 0$ (eq. 3), using heat capacities $\frac{9}{2}R$ and $\frac{7}{2}R$ for adsorbed CO_2 and N_2 and a fitted sorbent heat capacity $C_{V,\text{sorb}}(T)$. The Julia code differs from eq. 4 as typeset in two places: it *adds* the desorption work term ($+\,d\,RT_{\text{mid}}$ where eq. 4 prints $-RT\,n$), and it takes the absolute value $|nRT\ln(P_i/P_{i-1})|$ of the pressure term. The published numbers follow the code (Table 2 is reproduced only with the code's sign; with eq. 4's sign BEVQID's nominal ξ would be 9.59 instead of 9.15 μmol/J, the auditor's figure), and our ledger keeps the code's convention. Eq. 5: $E_3 = 0$ ("we assume no energy recovery"; the text adds that real systems would recover some). Eq. 9 is $C = 1/\xi$. Eq. 10 is the optimisation prior: $T_1 \sim U(200, 400)$ K (the optimiser bound in `Path_optimizer.jl` is also $[200, 400]$ K), $\Delta T \sim U(0, 200)$, $P_1 \sim U(101325,\ 1.1 \times 101325)$ Pa, $\Delta P \sim U(-101225, 0)$ Pa. Our Python re-implementation follows the Julia functions line by line (`data/code/k1/dac_model.py`); the Julia code is not redistributed here (Section 11).

Step 3: why heats at face value differ from exergy. [L] [C] A joule of heat is not a joule of work. Heat Q delivered at temperature $T > T_0$ (with T_0 the ambient temperature) is worth at most $Q(1 - T_0/T)$ of work, the Carnot factor; this is the cost of producing that heat reversibly from work. Removing heat Q *below* ambient (e.g. cooling the sorbent to $T_1 \approx 200$ K while it adsorbs) needs a refrigerator, which costs at least $Q(T_0/T - 1)$ of work; heat rejected above ambient can be thrown away (no-credit accounting, $r = 0$) or partly turned back into work (full reversible recovery, $r = 1$). The paper's face-value sum treats all three alike and sets the return step to zero, which is the point of our audit. We therefore (i) fix an explicit $T_0 \in \{273.15, 298.15, 313.15\}$ K; (ii) record every heat of the *sorbent and its adsorbate* once, with its direction and temperature (Step 1: Q_{ads} removed at T_1; Step 2: the per-step heat at the mid-step temperature; Step 3: sensible cooling of the sorbent and of the adsorbed gas from T_{end} back to T_1, an *explicit return step*, discretised with the same right-point rule $c_i\,\Delta T_i$ that the published Step 2 uses, so that the sorbent's heating and cooling close exactly; v1 used a trapezoid here, see Section 3.6 and the errata); (iii) weight each by $|1 - T_0/T|$ with the sign convention above; (iv) keep the works as published. The cost is $B(r) = B_{\text{NC}} - r \cdot (\text{credits})$ per mole of CO_2 and $\xi_{\text{ex}} = 1/B$. **Two assumptions of this "base" accounting carry weight** (audit item B3): the *contacted feed air* is not in the ledger (adsorbing at $T_1 < T_0$ means the air passing the sorbent must be at T_1 too, about 2,500 mol per mol CO_2), and the machines are ideal (Carnot refrigerator and heat pump). Section 3.5 prices the feed air and degrades the machines. Lean states the two weights (`carnot_in_mem_unit`: $0 < 1 - T_0/T < 1$ for $T > T_0$, `DAC-A-0001`; `refrig_weight_lt_one`: $0 < T_0/T - 1 < 1$ for $T_0/2 < T < T_0$) [A]; that these are the right weights is the textbook argument, not a Lean statement.

Step 4: the reversible floor. [A] [X] Even a perfect machine must pay the ideal-mixture minimum work to extract CO_2 from the mixture. For a pure product at mole fraction x this is $W_{\min} = RT_0\ln(1/x)$, which is 19.4 kJ/mol at $x = 400$ ppm and $T_0 = 298.15$ K (hand arithmetic; $1/W_{\min} \approx 51.6$ μmol/J). Lean proves that this function is positive for $0 < x < 1$ and strictly decreasing in x (`wmin_pure_pos`, `wmin_pure_strictAnti`, `DAC-A-0004`); for less pure products our code uses the corresponding mixture expression (Tier X, not in Lean).

A model whose reversible exergy cost B_{rev} (all credits taken) falls *below* $W_{\min}$ would violate the second law, so it would signal inconsistent inputs, provided our own Step-3 quadrature is consistent with the published Step 2 (v1's was not; Section 3.6). This is H2.

Step 5: the closed-form thresholds in a stylised surrogate. [C] To see *why* a cold start can still win, we use a two-component surrogate of the cycle: van 't Hoff Henry loadings for CO_2 and N_2 with heats of adsorption q_C, q_N, a sensible-heat term $c = \gamma R n_0$ and a recovery fraction r on the sensible heat, started at $T_c < T_0$ instead of T_0 and heated to the same T_h. The benefit is

$$\beta = C(T_0) - C(T_c) = A + \lambda B + (1 - r)\,\gamma R\,D, \qquad A_{\mathrm{ex}} = -(T_0 - T_c)\Big(\frac{q_C}{T_c} - R\Big), \quad A_{\mathrm{fv}} = R(T_0 - T_c),$$

where λ is the N_2-to-CO_2 loading ratio at the inlet and B, D are differences of loading-ratio-weighted terms (`DAC-C-0001`; derivation by hand, algebra in Lean). Beside each step the Lean statement is:

- the CO_2 term alone penalises cold under exergy: $A_{\mathrm{ex}} < 0$ whenever $q_C > RT_c$ (`numerator_factor`, `co2_term_penalises_cold`, `DAC-A-0002`); at face value it favours cold ($A_{\mathrm{fv}} > 0$, `co2_term_facevalue_favours_cold`). This is *not* the sign of the whole benefit;
- β is affine in r (`benefit_affine_in_r`), and when $\gamma D \geq 0$ the case $r = 1$ is the worst case for a cold start (`worst_case_full_recovery`, `DAC-A-0003`); the sign of D is *not* certified over the box (`cert_sign.log`: $D_{\mathrm{ex}} > 0$ certified on 2.9% of it, 97.1% undecided), so "$r = 1$ is the worst case" is conditional;
- **N_2 threshold** (at $r = 1$): for $B > 0$, cold wins iff $\lambda > \lambda^* = -A_{\mathrm{ex}}/B = (T_0 - T_c)(q_C/T_c - R)/B$ (`lambda_threshold_iff`);
- **heat-capacity threshold** (for $r < 1$): with $N = -(A + \lambda B)$ and RD replaced by $(1 - r)RD > 0$, cold wins iff $\gamma > N/((1 - r)RD)$ (`threshold_iff`);
- a rational instance ($R = 8.314$, $q = 40$ kJ/mol, $T_c = 200$ K, $T_0 = 298.15$ K) with negative CO_2-only term, checked by `instance_r1` (this is `norm_num`, an identity check and not a theorem about the world).

Reading: under exergy accounting the refrigeration of the CO_2 adsorption heat is a real penalty, and a cold start wins only because lowering T changes the *ratio* of N_2 to CO_2 uptake (the N_2 term λB and the sensible term). This is the paper's own mechanism (CO_2 uptake diverges from N_2 at low T); the purity at the optimum is 0.98 at 200 K but 0.20 at 240 K for LETQAE01 on its slice. At $r = 1$ this surrogate predicts the wrong sign for most Table 2 rows (Section 3.8), so the mechanism is *illustrated*, not established.

3 Results

3.1 Positive control: Table 2 reproduced before any re-weighting

[X] (`DAC-X-0001`) The Python re-implementation, fed the public dataset, reproduces the 14 rows of Table 2 in all eight printed fields each (14/14 rows, from the stored posterior means and standard deviations, which round to the printed strings), and recomputes every stored nominal array (loadings, heats, every Step-1/Step-2 array, E_1, E_2, E_{total}) with a maximum normalised deviation of 3.8×10^{-11} (one number per material, the density, is back-solved from one stored Henry constant because the CSD material files are not in the public record, see Section 6). The posterior statistics cannot be regenerated bit for bit (unseeded draws, Section 6); the nominal ξ differs from the posterior mean by -2.3% to $+1.3\%$. The Gaussian-process heat capacity re-fit agrees with the stored one to 4.0×10^{-6} relative. A second, vectorised implementation (`fast_costs.py`) agrees with the ledger evaluator to 2.3×10^{-15}.

3.2 Exergy re-accounting of the 14 published paths

Three readings, each [X] and each inside the published model. (1) *Exergy-weighted efficiencies are higher than face value*, not lower: $\xi_{\mathrm{NC}} = 25.4$–36.8 μmol/J against $\xi_{\mathrm{fv}} = 5.98$–11.26 (nominal; printed 6.12–11.259). The reason is *not* that sorbent heating dominates (v1 said so; corrected): on the 14 stored paths the sorbent heating is only 1–45% of the face-value E_{total}, while the heats of adsorption (Step 1) plus desorption (Step 2) are 48–89%

Table 1: The 14 Table 2 rows, base accounting (feed air not priced; [X]; `DAC-X-0002`, `DAC-X-0003`, `DAC-X-0004`; v2 numbers, Step-3 right-point rule, v2 grid). ξ_{pub}: printed posterior mean (Table 2). ξ_{fv}: our nominal recomputation at face value. ξ_{NC} ($r = 0$) and $\xi_{r=1}$: exergy-weighted ξ on the *stored* path at $T_0 = 298.15$ K, in μmol/J. Refrig.: refrigeration share of the no-credit exergy cost. T_1^*: exergy-optimal start ($r = 0$, $T_0 = 298.15$ K, unconstrained joint grid). fv, r=0, r=1: cold-start benefit ratio (best ξ with $T_1 < 298.15$ K over best ξ with $T_1 \geq 298.15$ K). Last column: highest purity of any warm path ($T_1 \geq 298.15$ K) on our grid / the stored purity used as the constraint; warm side empty under the constraint in every row.

material	ξ_{pub}	ξ_{fv}	ξ_{NC}	$\xi_{r=1}$	refrig.	T_1^* (K)	fv	r=0	r=1	warm/stored purity
VANNIK	11.259(38)	11.26	36.5	38.9	0.69	200.0	30.6	11.9	5.33	0.08/0.939
VANNIK	11.223(41)	11.24	36.6	38.9	0.69	200.0	29.1	11.5	5.32	0.08/0.939
LETQAE01	11.22(18)	11.21	36.8	38.5	0.76	200.0	16.4	10.2	6.59	0.06/0.983
BEVQID	9.186(66)	9.15	34.3	39.1	0.67	210.4	78.1	16.7	1.51	0.68/0.993
BEVQID	9.156(68)	9.12	34.8	39.2	0.66	210.1	82.8	18.2	1.51	0.68/0.994
FASJAL	8.186(69)	8.19	35.0	37.0	0.47	236.5	10.2	3.52	1.29	0.92/0.995
FASJAL	8.148(62)	8.14	35.1	37.7	0.48	235.0	12.2	4.14	1.29	0.92/0.996
FASJAL	8.028(69)	8.03	34.5	37.6	0.47	234.4	13.2	4.47	1.29	0.92/0.996
IFUDAO	7.26(29)	7.27	31.3	38.5	0.68	217.5	121.5	24.5	1.33	0.97/1.000
IFUDAO	7.00(28)	7.03	31.5	38.9	0.64	217.7	117.5	23.3	1.33	0.97/1.000
MAXHEA	6.84(36)	6.77	28.2	38.6	0.72	213.1	209.3	35.9	1.34	0.99/1.000
MAXHOK	6.56(41)	6.64	28.4	39.1	0.74	213.0	289.5	55.8	1.34	0.99/1.000
MAXHOK	6.26(52)	6.26	29.0	39.2	0.71	213.3	254.6	46.9	1.34	0.99/1.000
PARFOF	6.12(52)	5.98	25.4	40.0	0.80	203.4	817.8	129.1	1.39	0.99/1.000

(`rev/table1_compare.json`; the auditor found 49–88% on four rows). Face value prices the Step-1 heat removal at weight 1, against a Carnot refrigeration weight $T_0/T_1 - 1 = 0.26$–0.49 at $T_1 = 200$–237 K, and the desorption heat at weight 1, against $1 - T_0/T \approx 0.05$–0.25 at 315–390 K. For $T_1 > T_0/2$ the published face-value E_1 therefore already *over*-prices sub-ambient heat removal relative to Carnot (this is what `refrig_weight_lt_one` states); the only sub-ambient cost the face value omits is the Step-3 return (and, Section 3.5, the feed air). The face-value ξ is an accounting-convention quantity and is *not* an upper bound on CO_2 per unit of work; the source says as much (pp. 14–15: heat recovery "could therefore exceed our stated upper limit"). (2) The reversible floor on the stored paths is $B_{\mathrm{rev}} = 25.0$–27.0 kJ/mol against $W_{\min}(P_0, y) = 18.8$–19.4 kJ/mol, i.e. $\xi_{\mathrm{rev}} = 37.0$–$40.0$ μmol/J, below $1/W_{\min}$ in all 14 rows. (3) Refrigeration is 47–80% of the no-credit exergy cost: the penalty the hypothesis relied on is real and large.

3.3 H1 in the base accounting: the optimal start temperature does not move up

[X] (`DAC-X-0003`) *Grid (v2, a choice).* Joint re-optimisation (Section 6 for the protocol and its amendment) over $T_1 \in \{200, 210, \ldots, 400\} \cup \{205, \ldots, 245\} \cup \{270, 272, \ldots, 298\} \cup \{273.15, 298.15, 313.15\}$ K, ΔT in 10 K steps with $T_{\mathrm{end}} \leq 400$ K (the C_V training range), $P_{\mathrm{end}} \in \{100, 10^3, 5 \times 10^3, 2 \times 10^4, 6 \times 10^4, P_1\}$ Pa, P_1 as stored; then a refinement ($T_1 \pm 10$ K in 2 K steps, $\Delta T \pm 10$ K in 5 K steps, $P_{\mathrm{end}} \times \{0.5, 0.7, 1, 1.4, 2\}$ clipped to $[100, P_1]$) around the best overall, best cold, best warm and best purity-constrained cell of 12 accountings; 4,720–5,902 paths per row. No-swing paths and paths with $\Delta n_{CO_2} \leq 10^{-6}$ mol/kg are excluded. The published optimiser allows $T_1 \leq 400$ K and $\Delta T \leq 200$ K, i.e. T_{end} up to 600 K; our cap $T_{\mathrm{end}} \leq 400$ K is a restriction (the auditor's wider domain on three rows changed none of its thresholds by more than 0.005). v1 used $T_1 \leq 320$ K and described its pressure range as "$[100, 60000]$ Pa"; its refinement actually reached $P_{\mathrm{end}} = 84{,}000$ Pa, i.e. the range was $[100, P_1]$.

Result. In **14 of 14 rows** the exergy-optimal T_1^* lies in $[200, 236.75]$ K for every $T_0 \in \{273.15, 298.15, 313.15\}$ K and every $r \in \{0, 0.5, 1\}$; 0 of 14 rows have $T_1^* \geq T_0$ in any variant, against the preregistered criterion (> 50% at $T_0 = 298.15$ K, $r = 0$). **Vacuous variants (audit item B4, all 14 rows checked):** with the purity constraint on, *no* warm path ($T_1 \geq T_0$) on our grid reaches the stored purity for any row at any T_0 (best warm purities 0.06–0.99 against stored 0.939–0.99999, last column of Table 1), so the 9 constrained variants cannot fail: "$T_1^* < T_0$" holds there by construction. The robustness claim of v1 ("14/14 in every one of 18 variants") is therefore restated as **14/14 in each of the 9 informative (unconstrained) variants, i.e. 126 of 252 row-variants; the other 126 are vacuous.** This is a statement on our grid. The face-value control puts all 14 optima below 273 K (maximum 232.1 K), as it must for the evaluator to be

trusted. What *does* move is the hot end: unconstrained optimal T_{end} is 230–400 K at face value and 301–340 K under no-credit exergy accounting (`DAC-X-0004`), because heating near ambient is cheap in exergy. The warm alternative is the best path with $T_1 \geq 298.15$ K: its purity collapses for some rows (VANNIK 0.055, LETQAE01 0.050 at $r = 0$) but not for others (IFUDAO, MAXHEA, MAXHOK, PARFOF 0.93–0.98).

3.4 The cold-start benefit shrinks but stays, in the base accounting

[X] (`DAC-X-0004`) In the ratio columns of Table 1 the benefit falls from 10–818 at face value to 3.5–129 (exergy, $r = 0$), 2.6–79 ($r = 0.5$) and 1.29–6.6 ($r = 1$). At full reversible recovery several rows approach 1 (FASJAL 1.29). Within the base accounting the conclusion is not overturned. It *is* overturned at $r = 1$ for most rows once the feed air is priced (next subsection). We did not test any process-level recovery.

3.5 New in v2: pricing the chilling of the feed air

[X] (`DAC-X-0013`) *The omitted term.* Adsorbing at $T_1 < T_0$ means the contacted air must be at T_1. In the model the inlet CO_2 mole fraction is $x = 400/10^6$ (`run_optimize_intrinsic_dac.jl`, line 15), so at capture fraction $f = 1$ (every CO_2 molecule of the contacted air captured; the case most favourable to a cold start) the air contacted per cycle is $n_{\text{air}} = \Delta n_{CO_2}/x = 2{,}500\,\Delta n_{CO_2}$. Bringing it from T_0 to T_1 with an ideal refrigerator costs $n_{\text{air}} c_p\,[T_0 \ln(T_0/T_1) - (T_0 - T_1)]$ ($c_p = \frac{7}{2}R$; for $T_1 > T_0$, the heat-pump analogue $n_{\text{air}} c_p[(T_1 - T_0) - T_0 \ln(T_1/T_0)]$). A recuperator of effectiveness $\varepsilon \in [0, 1]$ recovers the share ε of it; we add $(1 - \varepsilon)$ times this exergy to $B(r)$. $\varepsilon = 1$ is the base accounting. For $f < 1$, replace $(1 - \varepsilon)$ by $(1 - \varepsilon)/f$. Note that $r = 1$ with $\varepsilon < 1$ is a mixed bookend: reversible for the sorbent's heats, irreversible for the air. The scenario family is computed on the same stored per-path components as the base accounting (`code/k1/rev_scan.py`, `rev_analyse.py`; the components reproduce the independent `fast_costs.py` to 1.0×10^{-15}).

Results at $T_0 = 298.15$ K, $\varepsilon = 0$, ideal machines (Table 2):

- $r = 0$ (the preregistered setting): a cold start still wins in **14/14** rows, but T_1^* moves from 200–237 K to 254–288 **K** (FASJAL 288, BEVQID 280–281, VANNIK 279–280, MAXHOK 273–274, PARFOF 254), and the benefit ratio falls to 1.46–8.0. **0/14 rows have $T_1^* \geq T_0$: H1 as preregistered is refuted with the feed air priced too.**
- $r = 0.5$: $T_1^* = 268$–290 K; FASJAL's three rows start only 8 K below ambient.
- $r = 1$: **for 11 of 14 rows the optimum moves to ambient.** On the coarse grid T_1^* is the 298.0 K cell; a refinement in 0.25 K steps over 294–298.15 K (`rev_refine_ambient.py`) gives $T_1^* = 297.25$ K (BEVQID) and 297.75 K (the others), and the best start below ambient beats the best start at or above it by 0.07–0.35%, less than each row's own printed ξ uncertainty (0.7–8.5%; a marginal posterior spread, so this is not a paired test): a tie at the model's own precision. Only VANNIK (both rows, 288 K, ratio 1.21) and LETQAE01 (284 K, ratio 1.36) keep a cold start, and they keep it for every $\varepsilon \in [0, 1]$.

Recuperator thresholds. How much of the air's cold must be recovered for a *materially* cold start to win depends on what "cold" is taken to be, so we report two splits (Table 2). With "cold" $= T_1 \leq T_0 - 10$ K against everything warmer, a cold start loses at $r = 1$ unless $\varepsilon > \varepsilon^* = 0.91$–$0.96$ (BEVQID 0.91, FASJAL 0.95, IFUDAO, MAXHEA, MAXHOK, PARFOF 0.95–0.96); on the auditor's split ($T_1 \leq 290$ K against $T_1 \geq 298.15$ K, its 10 K grid) $\varepsilon^* = 0.77$ (BEVQID), 0.88 (FASJAL), 0.89–0.90 (IFUDAO, MAXHEA, MAXHOK, PARFOF). At $r = 0.5$ only FASJAL has a threshold (0.12–0.37, "cold" $\leq T_0 - 10$ K); at $r = 0$ none. With the strict v1 split ($T_1 < T_0$ against $T_1 \geq T_0$) there is no crossing at all, because a start a fraction of a kelvin below ambient is always available; this is the tie described above, not a cold-start benefit. *Other ambients* ($\varepsilon = 0$, $r = 1$, coarse grid): the same 11 rows start at ambient to within 2 K at $T_0 = 273.15$ and 313.15 K as well (9 exactly at T_0, where the best start below ambient *loses* by 0.2–0.6%; BEVQID at 272 resp. 311.15 K, winning by $\leq 0.7\%$), while VANNIK and LETQAE01 stay 4–19 K below ambient. Read literally, $T_1^* \geq T_0$ holds in 9/14 rows at each of these two ambients and in 0/14 at 298.15 K; the difference reflects only where the grid cell next to ambient lies (1.15, 3.15 and 0.15 K below it). At $r = 0$ the cold start wins in 14/14 rows at 313.15 K and in 10/14 at 273.15 K (4 near-ambient).

Machine efficiencies (sensitivity only; `scenario_summary.json`). *Refrigerator degraded, heat pump ideal* ($\eta_R < 1$, a mixed bookend): without feed air, at $r = 1$ a start ≥ 10 K below ambient loses below $\eta_R^* = 0.10$–0.75 (VANNIK 0.14, LETQAE01 0.10, BEVQID 0.61, FASJAL 0.69, others 0.72–0.75); at $r = 0$ the auditor-split

threshold is $\eta_R^* = 0.012$–0.077 (none for PARFOF). With the feed air at $\varepsilon = 0$ and $r = 0$, a cold start loses below $\eta_R^* = 0.19$–0.93. *Both machines degraded symmetrically* (heat-pump and refrigeration costs divided by η, credits multiplied by η): without feed air the cold start is *strengthened* in 14/14 rows for every $\eta \in [0.01, 1]$ (at $r = 1$ the ratio at $\eta = 0.01$ is 1.7–62), confirming the auditor; with the feed air priced it is not uniform (at $r = 0$ FASJAL loses below $\eta = 0.89$, 0.10 or never on its three rows; at $r = 1$ several rows regain a cold start below $\eta \approx 0.95$–0.99). We do not summarise the priced-air, degraded-machine cases as either direction.

Comparison with the auditor's own numbers (three overlapping rows, FASJAL 8.186, BEVQID 9.186, VANNIK 11.259; `analysis.json`): T_1^* at $r = 0$, air priced: 290/280/280 K (auditor) against 288.15/281.15/279.15 K (ours), within one grid cell; η_R^* at $r = 1$ without air: 0.67/0.59/0.13 against 0.68/0.60/0.14 (within 0.015); η_R^* at $r = 0$: 0.055/0.031/0.037 against 0.077/0.039/0.050; ε^* at $r = 1$ on the auditor split: 0.84/0.68/none against 0.88/0.77/none (+0.04, +0.09); base benefit ratios at face value and $r = 0$ are 9–19% lower in our scan, at $r = 1$ within 3%; vacuity 0 warm cells meeting the stored purity in both. Plausible causes of the differences: the auditor interpolated the stored C_V instead of re-fitting the GP, allowed T_{end} up to 600 K and used a different grid without refinement. **The revised numbers have been checked by the producer, and the auditor's numbers agree where they overlap within these tolerances; an independent re-check of the new scenario family as a whole is pending.**

Table 2: Feed-air scenario family at $T_0 = 298.15$ K, $\varepsilon = 0$ (no recuperation), ideal machines, unconstrained ([X]; `DAC-X-0013`). T_1^* with outcome C (cold, $\leq T_0 - 10$ K), N (near-ambient, within 10 K below T_0) or W (warm). q_{10}: best ξ with $T_1 \leq T_0 - 10$ K over best ξ with $T_1 > T_0 - 10$ K. q_{v1} ($r = 1$): best ξ with $T_1 < T_0$ over best ξ with $T_1 \geq T_0$ (coarse grid; refined values in the text). ε_{10}^*, $\varepsilon_{\text{aud}}^*$: recuperator effectiveness below which a cold start loses at $r = 1$, for "cold" $= T_1 \leq T_0 - 10$ K, resp. the auditor's split ($T_1 \leq 290$ against $T_1 \geq 298.15$ K); none = cold wins for all ε. η_R^*: refrigerator efficiency below which a start $\leq T_0 - 10$ K loses at $r = 1$ without feed air.

material	ξ_{pub}	T_1^*, r=0	q_{10}	T_1^*, r=0.5	q_{10}	T_1^*, r=1	q_{v1}	q_{10}	ε_{10}^*	$\varepsilon_{\text{aud}}^*$	η_R^*
VANNIK	11.259(38)	279.15 C	1.27	281.15 C	1.17	288 C	1.2137	1.01	none	none	0.14
VANNIK	11.223(41)	280 C	1.25	281.15 C	1.16	288 C	1.2120	1.01	none	none	0.14
LETQAE01	11.22(18)	280 C	1.23	281.15 C	1.16	284 C	1.3648	1.05	none	none	0.10
BEVQID	9.186(66)	281.15 C	1.37	283.15 C	1.19	298 N	1.0010	0.78	0.91	0.77	0.60
BEVQID	9.156(68)	280 C	1.42	283.15 C	1.22	298 N	1.0010	0.78	0.91	0.77	0.61
FASJAL	8.186(69)	288.15 C	1.01	290.15 N	0.97	298 N	1.0005	0.74	0.95	0.88	0.69
FASJAL	8.148(62)	288 C	1.02	290 N	0.99	298 N	1.0005	0.74	0.95	0.88	0.69
FASJAL	8.028(69)	288 C	1.03	290 N	0.99	298 N	1.0005	0.74	0.95	0.88	0.69
IFUDAO	7.26(29)	280 C	1.49	282 C	1.29	298 N	1.0004	0.73	0.95	0.89	0.72
IFUDAO	7.00(28)	280 C	1.46	282 C	1.27	298 N	1.0004	0.73	0.95	0.89	0.72
MAXHEA	6.84(36)	278 C	1.58	280 C	1.37	298 N	1.0004	0.73	0.95	0.90	0.73
MAXHOK	6.56(41)	273.15 C	2.07	276 C	1.72	298 N	1.0004	0.73	0.96	0.90	0.74
MAXHOK	6.26(52)	274 C	1.78	279.15 C	1.52	298 N	1.0004	0.73	0.96	0.90	0.74
PARFOF	6.12(52)	254 C	2.70	268 C	2.12	298 N	1.0004	0.74	0.95	0.89	0.75

3.6 H2: no published ξ above $1/W_{\text{min}}$; no second-law violation in the dataset

[X] (`DAC-X-0005`) **The preregistered part stands.** Over the 8,765 materials with valid paths (15,161 paths; the public record holds 11,656 optimised files) the largest published face-value ξ is 11.259 μmol/J (VANNIK), against $1/W_{\text{min}} \approx 51.6$ μmol/J: no published ξ exceeds $1/W_{\text{min}}$.

Retraction (v2). Version 1 reported that five dataset entries (CSMNOX, GUSNEN, PACYIE, SABWAU01, UJOKIP05) had heat flows inconsistent with the second law, which it read as inconsistent inputs. **That claim is withdrawn: the apparent violations were produced by our own quadrature rule and vanish under the published rule.** Our G3 floor check first flagged 18 path/T_0 pairs at 298.15 K in five materials (14 at 273.15 K, 21 at 313.15 K, against the lowest-pressure floor), and 16, 22 and 25 pairs (63 in all, ten materials) against the P_0 floor. All of them come from the v1 Step-3 return, which we discretised with a trapezoid while the published Step 2 uses the right-point rule $c_i \Delta T_i$; the mismatch is of order $(dC_V/dT)\,\Delta T_{\text{step}}$ per step. Every flagged path has a tiny working capacity ($\Delta n_{\text{CO}_2} = 3 \times 10^{-9}$ to 3.4×10^{-4} mol/kg, face-value $\xi = 6 \times 10^{-8}$ to 2.9×10^{-3} μmol/J), so the mismatch, divided by Δn_{CO_2}, gave B_{rev} down to -1.5×10^6 kJ/mol. We reran the whole record with both rules (`code/k1/rev_h2_rules.py`, all 15,161 paths, both floors, all three T_0, with the v1 counter bug fixed): the trapezoid reproduces the v1 flags exactly (14/18/21 and 16/22/25),

and **the right-point rule gives 0 violations at every T_0 against both floors**. With the right-point rule the G2 sensible closure holds to 4.4×10^{-16} on every path (exact by construction). Two further errors of v1 are corrected with the retraction: SABWAU01's value was $B_{\text{rev}} = -41.9$ kJ/mol (v1 printed "J/mol"), and v1 called SABWAU01 and UJOKIP05 "not degenerate"; they have purities 0.91 and 0.67 but working capacities of only 4.2×10^{-5} and 1.4×10^{-4} mol/kg ($\xi_{\text{fv}} = 2.4 \times 10^{-4}$ and 8.8×10^{-4} μmol/J), so they are degenerate in working capacity. These flags were evidence about our floor, not about the dataset entries. Changing the Step-3 rule moves the Table 1 exergy numbers by at most 2.4×10^{-4} relative (all 14 rows, three T_0, three r); no printed digit of Table 1 changes, and the 14 rows still pass the gate. The v1 counter note (the first script skipped cases with $W_{\min}(\text{low } P) < 0$) is superseded by the full rerun.

3.7 Beyond the 14 rows: a seeded sample and the whole record

[X] (`DAC-X-0006`) A slice scan (T_{end}, P_1, P_{end} held at the stored values) on a seeded random sample of 150 materials (seed 20261027; each represented by its stored Pareto path of highest face-value ξ) finds $T_1^* \geq T_0$ in only **3 of 150** at $T_0 = 298.15$ K, $r = 0$. Re-selecting among all 15,161 stored Pareto paths, the no-credit-exergy-best path starts below 298.15 K for 8,420 of 8,765 materials (face value: 8,416). Both are conditional on the stored paths and are not re-optimisations; the sample is not a population estimate.

3.8 The surrogate: where it agrees with the model and where it does not

[C] (`DAC-C-0002`–`DAC-C-0004`) Over the box $T_c \in [200, 272]$, $T_0 \in [273, 313]$, $T_h \in [315, 400]$ K, $q_C \in [12, 67]$ kJ/mol, $A_{\text{ex}} < 0$ (certified; trivial since $q_C \geq 12$ kJ/mol exceeds $RT_c \leq 2.3$ kJ/mol). At the surrogate parameters extracted from 164 parameter sets (the 14 paths plus the sample), the certified signs of β at $T_0 = 298.15$ K, $r = 0$ are 151 positive (cold wins), 8 negative, 0 undecided (159 sets with $T_c < T_0$); at $r = 1$, 131 positive and 28 negative. Intervals are outward-rounded mpmath (80 bits; python-flint is not installed). An independent plain-float implementation lies inside the enclosures for 2854 of 2856 points; the other two differ by 3×10^{-8} relative from float cancellation, with the same sign. (The first surrogate run omitted the factor R on the sensible term; this float check caught it with 1904 of 2856 mismatches and the run was repeated.)

[X] (`DAC-X-0007`) The *link* between surrogate and model is empirical: the surrogate sign agrees with the full-model slice scan on 149/150 ($r = 0$), 146/150 ($r = 0.5$) and 128/150 ($r = 1$) of the sample at $T_0 = 298.15$ K. On the 14 Table 2 rows the surrogate is **wrong at $r = 1$ for most rows**: it disagrees on 11 of 14 at $T_0 = 273.15$ K, 10 of 13 scorable rows at 298.15 K and 5 of 8 at 313.15 K (it predicts that cold loses, the full model says cold wins); at $r = 0$ all scorable rows agree and at $r = 0.5$ one material (FASJAL) disagrees. (The count falls because rows with no grid point on one side of T_0 are not scorable; the figures come from a per-row recount, whereas the run's own agreement file uses the first path of each material.) The certified statements are therefore about the surrogate only, and the surrogate is a fair proxy of the published model only for $r \leq 0.5$. The slice comparison is also lopsided: with T_{end} held at 308–384 K, the best warm slice point has $\xi = 0.01$–0.4 μmol/J against 35–39 μmol/J cold, so the full-model "cold wins" on the slice is near-trivial (auditor's figures).

4 The headline, and its tiers

statement	tier	basis
the source model, eqs. 1–10, Table 2, licences	[L]	arXiv:2501.04825v1; Zenodo; NIST (`DAC-L-0001`, `DAC-L-0002`)
Carnot weights in $(0,1)$; $A_{\text{ex}} < 0$ for $q > RT_c$; affine in r; λ and γ threshold equivalences; $RT_0 \ln(1/x)$ positive, decreasing	[A]	`DacExergy/Core.lean`, 12 theorems (`DAC-A-0001..0004`)
decomposition $\beta = A + \lambda B + (1-r)\gamma RD$ and its thresholds in the surrogate	[C]	hand derivation + Lean (`DAC-C-0001`)
certified surrogate signs at 164 parameter sets; float cross-check	[C]	mpmath intervals (`DAC-C-0003`, `DAC-C-0004`)
Table 2 reproduced; exergy re-accounting; base joint optimum $T_1^* \in [200, 236.75]$ K (126 informative row-variants); benefit ratios; H2; sample; surrogate agreement; feed-air family; H1 negative control	[X]	grids and scans (`DAC-X-0001..0007`, `DAC-X-0013`, `DAC-X-0014`)
For the 14 published paths, in the published simplified model: the cold-start conclusion survives exergy accounting of the model's own heats and works (ideal machines, feed air not priced; 14/14 rows in all 9 informative variants). With the chilling of the feed air priced ($\varepsilon = 0$, $T_0 = 298.15$ K) a cold start still wins in 14/14 rows at $r = 0$ but at 254–288 K, and at $r = 1$ the optimum moves to ambient (within 0.9 K, benefit $\leq 0.35\%$) in 11/14 rows. H1 as preregistered ($T_0 = 298.15$ K, $r = 0$) is refuted in both accountings	[X]	`DAC-X-0011`, resting on `DAC-X-0001`, `DAC-X-0003`, `DAC-X-0004`, `DAC-X-0008`, `DAC-X-0013`, `DAC-X-0014`

The headline is filed Tier X because it rests on grid optimisation of the real model, not on a proof; the ledger gate would block it if it were filed above any claim it depends on. **Status of H1, by scenario** (row counts out of 14, unconstrained). Base accounting: refuted at every T_0 and r (0/14 with $T_1^* \geq T_0$; 14/14 start ≥ 10 K below ambient). Feed air priced, $\varepsilon = 0$, at the preregistered $T_0 = 298.15$ K, $r = 0$: refuted (0/14; 14/14 start ≥ 10 K below ambient). Outside the preregistered setting the priced scenarios do *not* refute it for most rows: at $r = 1$ the start moves to ambient in 11/14 rows (literally $T_1^* \geq T_0$ in 0/14 at 298.15 K but in 9/14 at 273.15 and at 313.15 K, Section 3.5), and a start ≥ 10 K below ambient loses in 3/14 rows already at $r = 0.5$ (FASJAL). With ideal machines, the cold start survives every feed-air pricing ($\varepsilon \in [0,1]$, every r) only for VANNIK (two rows) and LETQAE01; with a degraded refrigerator and the air priced even these lose (VANNIK below $\eta_R \approx 0.29$ at $r = 0$).

5 Controls

All controls were run before the exergy numbers were interpreted (`DAC-X-0008`, `DAC-X-0001`, `DAC-A-0005`).

Positive controls (all met).

- **Preregistered: Table 2 reproduced to the printed digits *from the stored posterior means*** before any re-weighting: 14/14 rows, 8/8 fields; nominal recomputation within $-2.3\% \ldots + 1.3\%$ (Section 3.1).
- **Preregistered: an all-$T \geq T_0$ path.** A path entirely at $T \geq T_0$ (300 K start) gives heat-in weights all in $(0,1)$, no refrigeration term, and heat-out free. This agrees with `carnot_in_mem_unit` [A] only as an identity: the control cannot fail.
- The real BEVQID ledger passes the consistency gate (G1–G4).
- The face-value joint optimum reproduces "start cold" (14/14 below 273 K); a stored path re-evaluated from raw inputs with the re-fitted heat capacity agrees to 4.2×10^{-7} relative.

Negative controls (all four rejected by the gate, as preregistered).

- **Double-counted sorbent heating**: rejected by G2 (the sensible closure between Step 2 and Step 3 is off by 100%).

- **Planted $\times 10^6$ unit error**: rejected by G4 (reported 9.1×10^6 against the ledger 9.15 μmol/J, and above $1/W_{\min} = 51.8$ μmol/J).
- **The sentence's value taken literally** (1.0×10^5 mol/J, i.e. 10^{11} μmol/J): rejected by G4.
- **Heats of adsorption removed** (a second-law violation planted): rejected by G3 ($B_{\text{rev}} = 8.7$ kJ/mol $< W_{\min} = 13.5$ kJ/mol).
- Lean planted controls (`DAC-A-0005`): `ctl_sorry` and `ctl_axiom` compile with rc 0 and are caught by `#print axioms` (`sorryAx`; `DacExergy.ctl_cheat`).

The four gate plants test the gate, not the H1 conclusion (audit item N12). **New in v2: a negative control for H1 itself** (`DAC-X-0014`). We plant an accounting in which a cold start must lose: every refrigeration cost (sorbent and feed air, $\varepsilon = 0$) is multiplied by M. A path with $T_1 \geq T_0$ rejects no heat below T_0, so its cost does not depend on M, while every path with $T_1 < T_0$ pays M times its Step-1 heat removal below ambient; for M large enough the warm side must win on any finite grid. Run through the same selection code (`rev_analyse.py`), $M = 10^4$ is detected in **14/14 rows at every** $r \in \{0, 0.5, 1\}$ ($T_1^* \geq T_0$, ratio < 1), so the H1 machinery can return a warm optimum. At $M = 100$ it is detected in only 7/14 rows at $r = 0$, 8/14 at $r = 0.5$ and 14/14 at $r = 1$: in the other rows the winner was the 298.0 K cell, whose refrigeration weight $T_0/T_1 - 1 = 5 \times 10^{-4}$ stays small even after multiplying by 100, which is the same near-ambient effect as in Section 3.5, not a failure to select. A sanity control in the other direction (refrigeration and feed air free) never makes a cold start worse than in the base accounting (14/14 rows, all r).

The gate (G1–G4) is ours: G1 every heat flow recorded once, G2 sensible-heat closure between Step 2 and Step 3, G3 $B_{\text{rev}} \geq W_{\min}$ at the delivered purity, G4 reported ξ equals $\Delta n / E_{\text{total}}$ in the stated units and lies below $1/W_{\min}$. It was applied to all 14 Table 2 rows (14 pass at $T_0 = 298.15$ K) and to the whole record in H2. G3 is meaningful only with a Step-3 quadrature consistent with Step 2 (v2: the right-point rule) and should be read together with a floor on Δn_{CO_2}: with the v1 trapezoid it produced the 63 spurious flags retracted in Section 3.6. All four negative controls were rerun under the v2 rule and are still rejected (double count: G2 off by 100%; heats removed: $B_{\text{rev}} = 8.73$ kJ/mol $< W_{\min} = 13.52$ kJ/mol).

The unit statement, stated neutrally. [L] [X] (`DAC-L-0003`, `DAC-X-0009`) On p. 20 the text reads "a few sorbents with intrinsic capture efficiencies above 1.0×10^5 mol/J". Table 2 gives ξ in μmol/J (largest 11.259); the dataset field `Intrinsic_capture_efficiency` carries the units string "mol/J" and its largest posterior value is 1.1259×10^{-5} mol/J; 60 of the 8,765 materials have a path above 1.0×10^{-5} mol/J. The next sentence compares with 2.46×10^{-6} mol/J (US grid), which is meaningful only at the 10^{-5} scale, and the exponent of that number prints with its minus sign in the same text extraction. The printed sentence is therefore consistent with 1.0×10^{-5} mol/J; the table, the data and the later sentences are self-consistent. We checked this against the dataset's units string and two text extractions of the PDF, *not* against the supplement (not obtained). The auditor inspected the page image: the typeset PDF (p. 20, rendered at 150 dpi) prints 1.0×10^5 mol/J with a positive exponent, while 2.46×10^{-6} on the next line shows its minus. So the positive exponent is in the typeset source; that 10^{-5} was meant remains an inference from the table, the data and the next sentence. We drew no conclusion from it other than the negative control above; it is not an error of the method and no author has been contacted.

6 Protocol, deviations and observations

Protocol amendment, written before the joint scan. The operational test (`logs/h1_operational_test_written_before_run.md`, written before any exergy number) scanned T_1 on a 1 K grid from 200 K while holding $(T_{\text{end}}, P_1, P_{\text{end}})$ at the stored values. After that slice scan, and before the joint scan, the protocol was amended (`logs/h1_protocol_amendment_2026-10-27T1300Z.md`): the slice measures the optimum along a slice that NSGA-III had chosen for a cold start, so warm starts with deeper vacuum and pure vacuum swings at $T_1 = T_{\text{end}} \geq T_0$ had never been evaluated. The amended score is the joint grid (coarse: $T_1 \in \{200, \ldots, 320\}$ K step 10, ΔT step 10 K with $T_{\text{end}} \leq 400$ K, $P_{\text{end}} \in \{100, 10^3, 5 \times 10^3, 2 \times 10^4, 6 \times 10^4\}$ Pa; refinement around the best cell of each accounting). The slice result is kept and reported as conditional (Section 3.6). We report both results, and the criterion ($> 50\%$ at $T_0 = 298.15$ K, $r = 0$) was not changed.

Deviations from the preregistration (the preregistered text of K1 is unedited).

1. **Material files missing.** The CSD material files (`cell_density`, `cell_mass`) are not in the public record. The density is back-solved from one stored Henry constant per material (every other array is an independent check); scans use the cell mass ρV_{box} from the saturation box for the saturation truncation. The location of the saturation edge (e.g. BEVQID's effective $T_1 \approx 209$–210 K) depends on this substitute; the direction of the H1 result does not.
2. **Table 2 reproduced from stored values**, plus an independent nominal recompute; the posterior itself is not regenerable (unseeded draws).
3. **Search space**: a 1-D slice, then a joint grid with refinement instead of NSGA-III over 8,759 materials (out of budget on 2 cores). The 150-material sample and the stored-Pareto re-selection are conditional. The first joint run let the refinement go to $P_{\mathrm{end}} = 50$ Pa, below the optimiser's 100 Pa bound; it was rerun with the bound enforced, conclusions identical (the first run is kept as `scan_joint_v1_Pend_unbounded`).
4. **No python-flint**: mpmath intervals instead. **Lean runner**: the pinned wrapper `lean_pinned.py`, not `anse/formal/lean_runner.py` (the repository's usual route); the rc and the axiom output were checked. The producer ran no `leanchecker` replay; the independent auditor recompiled `Core.lean` (rc 0) and replayed `Programmes.DacExergy.Core` with `leanchecker` (rc 0, 118 s), checking its own build of the unchanged source (sha256 as below; `data/leanchecker_replay_audit.json`).
5. **Supplement not read** (K1c required checking the sentence against it before any mention; we mention it anyway, as instructed by the author, on the evidence above). No author contact (the proposal reserves that decision to the owner).
6. **GATE-1** (the statement-adequacy and vacuity checks added by the proposal addendum) was not run. Several Lean theorems are one-line tactic identities (`ring`, `norm_num`); `instance_r1` and `benefit_affine_in_r` should be read as such, and a linter notes an unused hypothesis (`h0`) in `refrig_weight_lt_one`.
7. **Counts**: the preregistration assumed one 11,660-material set; the public record holds 11,656 optimised files, of which 8,765 have valid paths. The `h2_all.py` own-counter under-counted (Section 3.5).
8. **Shell discipline**: the author's isolated-shell rules (no variables, loops or heredocs) were lapsed in a few commands while preparing this paper (heredocs for several read-only inspection commands, one compound `cd`, one command with a shell variable that the guard refused); none touched git. During the v2 revision: about 13 commands were compound `cd X; ...` or `cd X && ...` commands (all read-only inspections except one, which ran the first `pdflatex` build and wrote its outputs), about 8 used `git -C` (read-only: `rev-parse`, `log`, `branch -r`, `show`, `ls-files`, `status`), several chained `echo`, `mkdir` or `cp` with `;`, several short inspections used `python -I -c` inline code rather than a script file, and a sleeping wait script ran outside the `taskset` core pin (no computation); no shell variable, loop or heredoc was used and no command changed git state.
9. **v2 grid**: the revision rescanned all 14 rows on a wider and finer grid (T_1 up to 400 K, 2 K steps from 270 to 298 K, $P_{\mathrm{end}} = P_1$ added) and with the closure-consistent Step 3; all Table 1 and ratio numbers of v2 come from this rescan, and the v1 scan files are kept for comparison.

Neutral observations on the published code and data [X] (`DAC-X-0010`); none is an error of the method, and none is a claim about the authors' intent.

- The posterior function draws 100 samples without a seed and, inside the loop, reuses one heat-capacity draw, so printed ξ uncertainties omit C_V variability and the means are not bit-reproducible; in the optimised JSON files `Captured_CO2` is the posterior mean while `E_Balance` is nominal, so they should not be divided by each other.
- Three descriptions of the single path of Fig. 4/Table 1: the text, 250→350 K then vacuum to 40,530 Pa; the stored SinglePath files and the dataset README, 250→375 K and 42,844.84 Pa (checked on BEVQID, LETQAE01, VANNIK); the code default, 300→350 K.
- Eight Table 2 rows have P_1 below 101,325 Pa (82,466–101,126 Pa) although eq. 10 and the v0.0.2 optimiser bound P_1 to between 101,325 and 111,457.5 Pa; the stored results were probably produced with other bounds or another code version.

- With the published evaluator, our joint grid finds nominal paths that exceed several Table 2 rows in both ξ and purity (PARFOF 8.42 against 5.98 μmol/J at purity 0.99999; MAXHEA 8.62 against 6.77; IFUDAO 8.60 against 7.27; nominal against nominal). The published front *may not be fully converged*; this is an observation to check, not a claim. Those face-value optima end below ambient (T_{end} 235–285 K), so their heating is cheap in exergy.

7 Lean formalisation

Modules `formal_cert/Programmes/DacExergy/` (copies in `data/lean/`): `Core.lean` (12 theorems, imports `Mathlib.Tactic` and `Mathlib.Analysis.SpecialFunctions.Log.Basic`; never `import Mathlib`) and `Controls.lean` (never imported). Compiled through the pinned wrapper (Lean 4.34.1), rc 0; every theorem's `#print axioms` reports exactly [`propext`, `Classical.choice`, `Quot.sound`] (`data/logs/lean_core.json`). No `sorry` or `native_decide`. The planted controls compile with rc 0 and the audit exposes `sorryAx` and `DacExergy.ctl_cheat` (`lean_controls.json`). sha256: `Core.lean` `4bfa4f7ebd979d53e8e98847369610339f5bc3530dbd7f43f983fbbfe1a6833c`; `Controls.lean` `25017505f1fd76b08024326b9debd03664179602272e025d1d4d1d4f0448aac8`. Independent check: the auditor recompiled `Core.lean` (rc 0, all twelve theorems on exactly the three allowlisted axioms), compiled `Controls.lean` (both plants exposed) and replayed `Programmes.DacExergy.Core` with `leanchecker`: rc 0 in 118 s (`data/leanchecker_replay_audit.json`); every hypothesis set in `Core.lean` was shown satisfiable. **What Tier A certifies:** twelve elementary real-algebra lemmas, three of them bare identities (`benefit_affine_in_r` and `numerator_factor` are `ring` identities, `instance_r1` is a `norm_num` numeric check); `lambda_threshold_iff` and `threshold_iff` are one-step linear rearrangements; `co2_term_facevalue_favours_cold` is a product of two positives; the rest are elementary facts (Carnot weights in $(0, 1)$, the refrigeration weight below 1 for $T > T_0/2$, the sign of A_{ex}, monotonicity in r, positivity and monotonicity of the logarithm). They certify no number in Section 3. **It does not certify** that the weights are the right physics, that the surrogate decomposition matches the published model, any grid optimum, or any number in Section 3; the transcendental parts (exponentials in the van 't Hoff loadings, logarithms in the sensible integral) are interval arithmetic outside the kernel (Tier C). Note that `co2_term_penalises_cold` is explicitly *not* the sign of the whole benefit.

8 Claim ledger

`ledger/build_ledger.py` files 26 claims: 5 at Tier A (`DAC-A-0001..0005`), 3 at L (`DAC-L-0001..0003`), 4 at C (`DAC-C-0001..0004`) and 14 at X (`DAC-X-0001..0014`; v2 added `DAC-X-0013`, the feed-air family, and `DAC-X-0014`, the H1 negative control, and rewrote `DAC-X-0005` as the retraction). The Tier C claims are phrased as statements about the surrogate at stated numeric parameter values, so they do not depend on the Tier X claims that describe how the values were extracted from the model; the link is the separate Tier X claim `DAC-X-0007`. Each claim has a content-addressed evidence blob in `ledger/evidence/`; the headline is `DAC-X-0011`; `DAC-X-0012` records the absences. The gate (`tools/ledger.py ledger.json --evidence-dir evidence`, verbatim in `ledger/gate_output.txt`) exits with code **1**: **0 blocking findings and 5 flags** (`LEDGER_UNAUDITED_TIER_A`, one for each Tier A row: no *person* has certified that the Lean statements mean what their glosses say). Exit code 1 is the honest state, not a failure to be cleared by editing. The independent AI audit is recorded on the five Tier A rows in an `ai_audit` field (auditor, verdict, report, kernel replay); the gate's `audit` field stays null because an AI audit is not a person. No statement lock. The v2 gate run (after the rebuild) gives the same result: exit code 1, 0 blocking, 5 flags.

9 Limitations

- **Everything is inside a published simplified model**: Henry/IAST loadings, absolute adsorption, adsorbed-gas heat capacities $\frac{9}{2}R$ and $\frac{7}{2}R$, GP-extrapolated heat capacity, binary CO_2/N_2 air, no water, ideal piston. A different model could give a different answer; nothing transfers to real plants.
- **Exergy weighting uses ideal reversible machines against one ambient** T_0 in the base accounting and, in v2, a feed-air term at capture fraction 1 with recuperator effectiveness ε; works are kept at face

value (dropping the PV terms or charging the product recompression does not change the direction in the base accounting, auditor's check); no process-level recovery was modelled; r, ε and η are bookends, not designs. **The size and, at $r = 1$, the existence of the cold-start benefit depend on the feed-air pricing** (Section 3.5); the degraded-machine cases are a sensitivity only.

- **The joint search is a grid with refinement, not a proof of global optimality** ($T_{\text{end}} \leq 400$ K, a restriction of the published domain); the cell-mass substitute shifts the saturation edge; the vacuity of the purity-constrained variants is a statement on this grid.
- **The interval results are about the surrogate** and the surrogate is wrong on 11 of 14 Table 2 rows at $r = 1$ and $T_0 = 273.15$ K (Section 3.7); the Lean results certify algebra only.
- **The supplement was not obtained and not read**; the sub-ambient pricing question is settled only for the code and the main text.
- **Audit status**: v1 was audited by an independent AI agent (not a person), verdict ACCEPT_WITH_CORRECTIONS (`AUDIT.md`); the auditor recompiled the Lean core and replayed it with `leanchecker` (rc 0, 118 s). This v2 is the producer's revision (`REVISION.md`); **the new feed-air scenario family, the H1 negative control and the full-record H2 rerun are not independently re-checked**; no human audit; no statement lock; GATE-1 not run. The advisor review of the K1 lane timed out and produced no output.
- **Repository guards were not run** (`antigravity_guard.py`, `test_rigor_guard.py`, `pytest tests/`); no guard output is claimed.
- **No author contact** and no statement about the authors' intent. Wording that implies the NIST bound is wrong is not supported and is not intended; the metric is a face-value convention by the source's own description.
- **Novelty is unchecked** beyond the source paper read; this is not a priority claim.
- **Data note**: the run note quotes "11/14" surrogate disagreement without a temperature; the per-row recount shows it holds at $T_0 = 273.15$ K, with 10/13 at 298.15 K.

10 Errata and revision record (v1 → v2)

The independent AI audit of v1 (`AUDIT.md`) raised four blocking items; the item-by-item response is in `REVISION.md`. In short:

1. **Retracted (our own claim):** "five dataset entries (CSMNOX, GUSNEN, PACYIE, SABWAU01, UJOKIP05) have heat flows inconsistent with the second law". The flags were an artefact of our Step-3 trapezoid; with the published right-point rule there are 0 violations over all 15,161 paths, every T_0, both floors (Section 3.6). Corrected with it: SABWAU01's B_{rev} is -41.9 kJ/mol (v1: "J/mol"), and SABWAU01 and UJOKIP05 are degenerate in working capacity.
2. **Corrected:** v1 said the paper's files were not redistributed and that nothing was pushed. Earlier commits of the public GitHub repository contained the arXiv text extraction, the NIST code tarball and tree, and 51 dataset files; the text and the NIST code were removed from the tracked tree on 2026-10-27, while the 51 CC-BY-4.0 dataset files are still tracked (Section 11). The Zenodo bundle of this paper contains none of them.
3. **Narrowed:** the title, abstract and headline. v1's "survives exergy accounting" held only for an accounting that omits the chilling of the feed air and assumes ideal machines; v2 adds the feed-air scenario family (Section 3.5), in which the cold start shrinks at $r = 0$ and disappears for 11/14 rows at $r = 1$.
4. **Corrected:** v1's robustness count "14/14 in every one of 18 variants" included 9 purity-constrained variants that are vacuous for every row; the informative count is 126 of 252 row-variants.
5. **Smaller corrections:** eq. 4 versus the code (sign of the desorption work, absolute value); eq. 10 prior $T_1 \sim U(200, 400)$ K; the v1 grid caps ($T_1 \leq 320$ K, $T_{\text{end}} \leq 400$ K) and the P_{end} actually reached (84,000 Pa); the reading "sorbent heating dominates" (it is 1–45%; the adsorption and desorption heats are 48–89%); the surrogate's $r = 1$ sign errors and the uncertified sign of D; the Tier A labelling (three

identities); stale audit and replay status lines; the unit-sentence page-image check; the arXiv licence in the bibliography; an H1 negative control added.

Numbers that changed: Table 1's ratios and T_1^* come from the v2 rescan (largest change: FASJAL's $r = 1$ ratio $1.2 \to 1.29$; T_1^* range 200–236.54 $\to$ 200–236.75 K); the stored-path exergy columns are unchanged at the printed digits (drift $\leq 2.4 \times 10^{-4}$).

11 Credit, licences and reproducibility

Credit. The model, the code, the dataset and Table 2 are the work of A. McDannald, D. W. Siderius, B. DeCost, K. Choudhary and D. L. Ortiz-Montalvo at the National Institute of Standards and Technology. We acknowledge NIST as the source of the data and software.

What the Zenodo bundle of this paper contains (and does not). The bundle (this paper's directory, `papers/p4_dac_exergy/`) contains **no third-party full text** (neither the arXiv PDF nor any text extracted from it; the arXiv licence of [1] is `arxiv.org/licenses/nonexclusive-distrib/1.0/`, which grants no redistribution right to third parties), **no NIST code tarball and no extracted NIST source tree**, and **no dataset files**. Third-party material is identified only by name and sha256: the 51 dataset files used for the Table 2 materials in `data/zenodo_members_SHA256SUMS.json` (CC-BY-4.0, available from Zenodo [2]), and the full archive `DAC_data.tar.gz` by sha256 `75a73c75a376a4611511af4a9e99b15af10096fa1886e5d969e4a1d8268df3fa` and md5 `a2520f520140c60f8a4663d5b403966c` (the Zenodo value). The licence notes are in Section 1 and the bibliography. *Our own material in the bundle*: the Python code (`data/code/k1/`), which re-implements the NIST Julia functions in Python. This code is a derivative of NIST-developed software: it was **changed** (translated from Julia to Python, with additional exergy accounting; revised on 2026-10-27 for the step-3 rule and the feed-air scenarios) by the author's AI agents; the two files derived from NIST sources (`dac_model.py`, `cv_gp.py`) carry a change notice in their headers, and the NIST licence notice is kept in `data/code/k1/NIST_NOTICE.txt`; the software is provided "as is" and neither NIST nor the authors warrant its correctness. We also include small logs of our own runs, the Lean sources, the run note, the audit and the revision note.

The public GitHub repository. The AutoevolveAI repository on GitHub is **public**. Earlier commits of its branch `worktree-openai-math-discovery` (the K1 lane commit `e29869853ec8056f6b8aaa1c32e09e018b80ae61`, pushed to `origin`) contained, under `results/proposal4/2026-10-27/K1/data/`, a text extraction of the arXiv paper, the NIST code tarball and its extracted source tree, and 51 dataset files. The arXiv text extraction and the NIST tarball and source tree were **removed from the tracked tree on 2026-10-27** (commit `b9a226cc1220be4e0006d0ae61ca5dd6b1a748a6`). **The 51 dataset files (with their `SHA256SUMS.json`) were not removed by that commit and are still tracked** under `results/proposal4/2026-10-27/K1/data/zenodo_members/` at the HEAD named below; they are CC-BY-4.0, which permits redistribution with attribution (given in Section 1 and [2]), and whether to untrack them as well is the owner's decision. All of these files remain in the repository's history, whose rewriting is also the owner's decision. We do not claim that nothing was pushed.

Reproducibility. Repository AutoevolveAI, branch `worktree-openai-math-discovery`; K1 lane committed at `e29869853ec8056f6b8aaa1c32e09e018b80ae61`; v1 of this paper at `92f8953c6a449366551d23b234839d99e650289a`; HEAD when this revision was written `754df91f5694d7001517083d97247b23214a67bd`, and the revision's files are *uncommitted* working-tree changes on top of it (not pushed, not published). The paper's `data/` holds the code, Lean sources, small logs, and the large logs `cert_sign.json` and `scan_T1.json`; the v2 logs (`h2_rules.json`, `analysis.json`, `scenario_summary.json`, `refine_ambient.json`, `exergy_table2_right.json`, `table1_compare.json`) are in `data/logs/rev/`; the large raw files stay in `results/proposal4/2026-10-27/K1/logs/`: v1 `h2_all.json` and `scan_joint_v1_Pend_unbounded.json` (sha256 `a3a59822...` and `3e615bf5...`), v2 `rev/scan_raw.json` (49 MB, sha256 `ca8376df...`, full value in `data/logs/rev/SHA256SUMS.json`). Commands (from the repository root):

```
python -I code/run_pc.py ; python -I code/run_exergy_table2.py          # positive control, exergy re-accounting
python -I code/run_scan.py ; python -I code/run_joint.py                # slice scan, joint grid (amended)
```

```
python -I code/h2_all.py ; python -I code/h2_violators.py               # H2
python -I code/surrogate_params.py ; python -I code/cert_sign.py        # surrogate, interval signs
python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/DacExergy/Core.lean
python -I data/code/table2_surrogate_check.py ... ; python -I data/code/recount_ranges.py ...   # paper-side recounts
python -I code/rev_h2_rules.py ; python -I code/run_exergy_table2.py     # v2: H2 both rules, full record; Table 1
python -I code/rev_table1_compare.py ; python -I code/rev_scan.py       # v2: drift and shares; component scan
python -I code/rev_analyse.py ; python -I code/rev_refine_ambient.py    # v2: scenario families, near-ambient refinement
python -I code/rev_summary.py ; python -I code/rev_tables.py            # v2: summary json and the two table bodies
python -I ledger/build_ledger.py ; python -I ledger/run_gate.py         # gate: exit 1 = 0 blocking, 5 flags
```

The exact argument lists are in the script headers and the run note. For v2 the H2 record pass (1,091 s), the Table 1 evaluation, the 14-row component scan (1,695 s) and the analyses were rerun; the v1 slice scan, sample and surrogate runs were not. The runs used two CPU cores (6–7) and no GPU.

References

[1] A. McDannald, D. W. Siderius, B. DeCost, K. Choudhary, D. L. Ortiz-Montalvo, *Intrinsic Direct Air Capture*, arXiv:2501.04825v1 (8 Jan 2025), National Institute of Standards and Technology. Licence: arXiv non-exclusive distribution licence 1.0 (`http://arxiv.org/licenses/nonexclusive-distrib/1.0/`); not redistributed here. Source of the model, eqs. 1–10 and Table 2 ([L]). Supplementary information not read.

[2] A. McDannald, D. W. Siderius, *Data for Intrinsic DAC calculations*, version 1.0, Zenodo, doi:10.5281/zenodo.14452152 (NIST Data Publication doi:10.18434/mds2-3687), CC-BY-4.0.

[3] NIST, *IntrinsicDACCycle* v0.0.2 (Julia package), `https://github.com/usnistgov/IntrinsicDACCycle`, NIST software licensing statement; read; not redistributed in this paper's bundle (its tarball was in the public repository's history, Section 11).

[4] The mathlib Community, *The Lean mathematical library*, arXiv:1910.09336 (2019); cited for the library only.